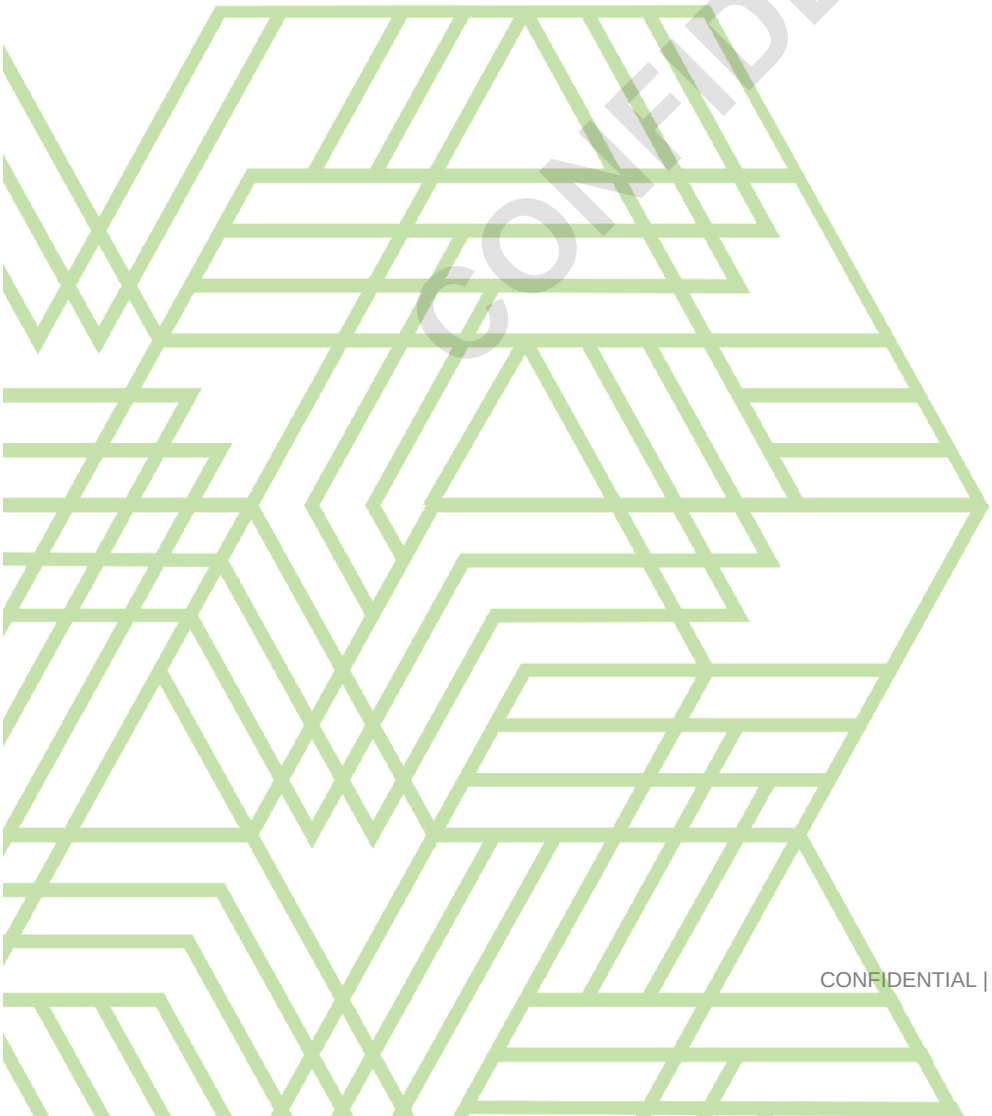
SEGGER Embedded Studio for ATM2/ATM3/ATM33 Wireless SoC Family

User Guide

SUMMARY: SEGGER Embedded Studio (SES) is an Integrated Development Environment with project management tools and editor. This document provides the installation and configuration required to use SES for Atmosic SDK.

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Table of Contents

1. Overview 3
1.1 Prerequisite 3
1.2 Limitation 3

2. ATM2/ATM3 Platform 4
2.1 Set Up Tool Chain 4
2.1.1 Set up the Atmosic SDK 4
2.1.2 Install SEGGER tools 4
2.2 Create SES application project 4
2.3 Compile application 5
2.4 Program firmware 5
2.4.1 Hardware setup 6
2.4.2 J-Link Flash Loader Setup 6
2.4.3 Download Firmware and Flash NVDS 7
2.5 Debug 8

3. ATM33 Platform 9
3.1 Installation 9
3.2 Generate SES Project File and NVDS 10
3.3 Build and Program Examples 10
3.4 Bootloader 11
3.5 Debugging 12

Revision History 13

List of Figures

Figure 1 - Build Output with No Errors
Figure 2 - Tool Chain Directory
Figure 3 - Atmosic Interface Board with J-Link Adapter Board
Figure 4 - Programming Flash Output
Figure 5 - SES Debug Session Windows
Figure 6 - ATM3330e/ATM3330 Evaluation Board
Figure 7 - SES IDE GUI - Compile
Figure 8 - SES IDE GUI - Program
Figure 9 - Build Bootloader
Figure 10 - Debug Mode

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1. Overview

This document provides the installation and configuration required to use SEGGER Embedded Studio (SES) for Atmosic SDK. SES is a cross platform Integrated Development Environment (IDE), which supports different operating systems.

1.1 Prerequisite

1) Atmosic SDK 5.1 or later
2) ATM33 platform
    a) ATM3330e/ATM3330 Evaluation Board (EVB)
3) ATM2/ATM3 platform
    a) Atmosic EVK and Interface Board
    b) SEGGER J-Link Plus debug probe with pin override support (Pin override is required for chip reset. Some J-Link models e.g. J-Trace don't support this feature and would get an error during target init process.)
4) SEGGER embedded studio (Confirmed version:V6.22a)
5) One of the following operating systems:
    a) Windows 10
    b) macOS
    c) Linux

1.2 Limitation

- Support gcc toolchain only:

  The SES projects work with the specific GNU gcc toolchain version (10.3-2021.07) only.

- Limited SDK functionality:

  The SES projects support the build and program bootloader, example firmware and NVDS data into ATMx devices via the J-Link. For the other SDK makefile functions (such as program OTP, pull firmware/NVDS/OTP, show NVDS/OTP data, etc.), please use MSYS2 console in the SDK. For ATM33, changing J-Link OB driver for openocd use might be required for certain functions.

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2. ATM2/ATM3 Platform

2.1 Set Up Tool Chain

2.1.1 Set up the Atmosic SDK

Follow the Atmosic SDK Quick Start Guide (available on the Atmosic support website) to set up the SDK. Enter the “make run_all” command in the command line to ensure that GNU/GCC can build an Atmosic SDK example, then download the example image into the Atmosic Evaluation Board.

2.1.2 Install SEGGER tools

Plug-in SEGGER J-Link debug probe into your laptop or PC and install the most recent release of the SES. Please make note of the SES installation destination path in the setup wizard. This path is required to put the J-Link flash loader file. Please also make sure the J-Link driver is installed correctly.

2.2 Create SES application project

- Linux and macOS
  1) Navigate to the examples folder (like: SDK_root\platform\atm2\ATM22xx-x1x\examples\BLE_adv)
  2) Enter the command in a command line to clean objects: make clean
  3) Enter the command in a command line to generate example code’s SES project according the chip: make ses_gen BOARD= m2201(or m2202, m2221, m2251, m3201, m3202, m3221, m3251)
  4) You will see a new folder named "ses". This is the SES project file for this example.
- Windows 10
  Each example will have a SES project (folder named "ses") after running the Atmosic Windows SDK installer. The default project is for m221. For other device, please generate project by command “make ses_gen” with BOARD type.

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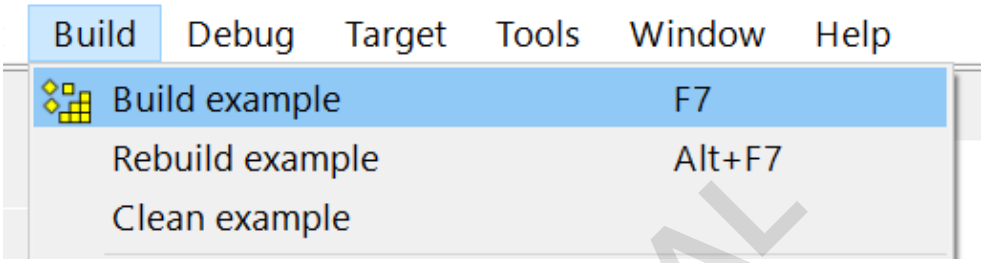
2.3 Compile application

Compile the application from a SES project

1. Open your project in SES

   SES projects are located in the sesauto subfolder of the example, such as **SDK_root\platform\atm2\ATM22xx-x1x\examples\BLE_adv\sesauto\example.emProject**

2. Select Build>Build example



Make sure that there are no build errors. The output should look similar to Figure 1:

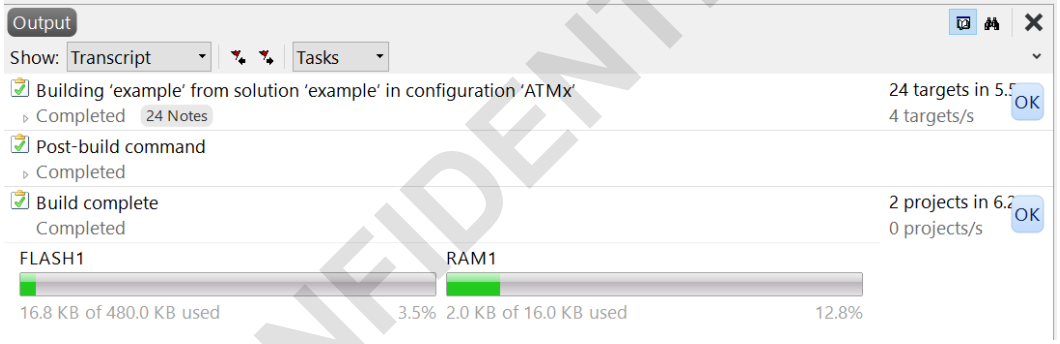


Figure 1 - Build Output with No Errors

The GNU/GCC toolchain will use Atmosic SDK's path for SES by default. If users want to modify the toolchain path, users need to modify it in SES ID, see Figure 2.

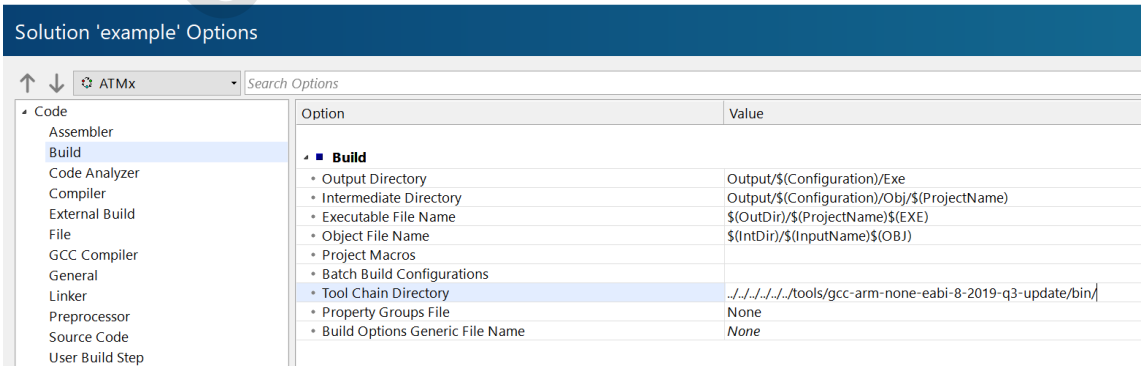


Figure 2 - Tool Chain Directory

Users also need to modify the post build script named postbuild.js in the sesauto folder.

2.4 Program firmware

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2.4.1 Hardware setup

Refer to the Keil MDK Configuration Guide for Atmosic SDK (available on the Atmosic support website) for details of the J-Link connection. Figure 3 shows the Atmosic interface board with J-Link adapter board. J-Link debug probe will not provide power to the EVB and still need to plug USB cable into USB1 of the EVK to supply the power to the EVB.

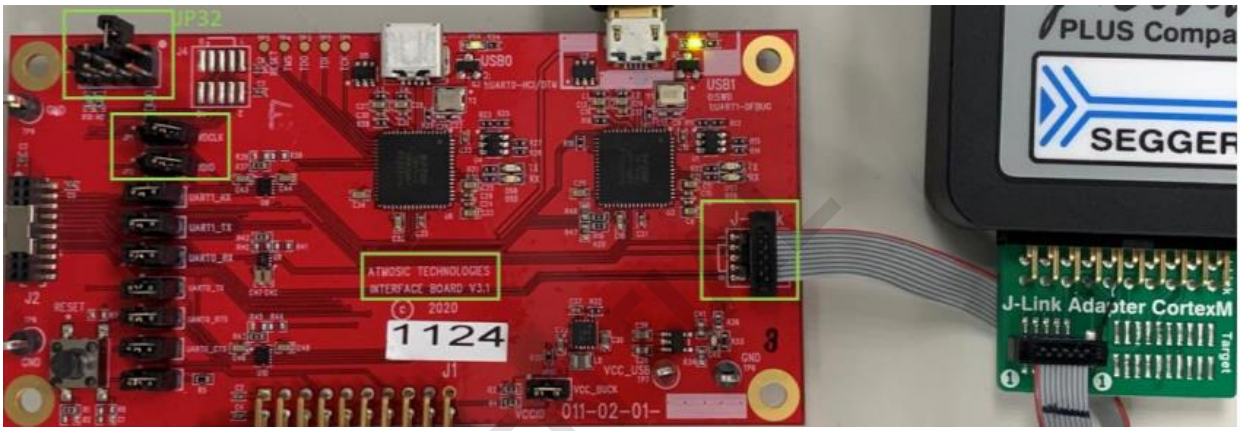


Figure 3 - Atmosic Interface Board with J-Link Adapter Board

2.4.2 J-Link Flash Loader Setup

There are three files (atmx_flash.FLM, atmx_nvds.FLM and JLinkDevices.xml) under SDK_root\tools\keil as flashloader. Please refer to the following to copy these files.

- Windows 10
  - Copy JLinkDevices.xml to C:\Program Files (x86)\SEGGER\SEGGER Embedded Studio for ARM xxxx\bin\
  - Create Atmosic folder under C:\Program Files (x86)\SEGGER\SEGGER Embedded Studio for ARM xxxx\bin\
  - Create atmx folder under C:\Program Files (x86)\SEGGER\SEGGER Embedded Studio for ARM xxxx\bin\Atmosic
  - Copy atmx_flash.FLM and atmx_nvds.FLM to C:\Program Files (x86)\SEGGER\SEGGER Embedded Studio for ARM xxxx\bin\Devices\Atmosic\atmx
- Linux
  - Copy JLinkDevices.xml to /usr/share/segger_emb…./bin
  - Create Atmosic folder under /usr/share/segger_emb…./bin
  - Create atmx folder under /usr/share/segger_emb…./bin/Atmosic

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- Copy atmx_flash.FLM and atmx_nvds.FLM to /usr/share/segger_emb…./bin/devices/Atmosic/atmx

- macOS
  - Copy JLinkDevices.xml to /Applications/SEGGER…./bin
  - Create Atmosic folder under /Applications/SEGGER…./bin
  - Create atmx folder under /Applications/SEGGER…./bin/Atmosic
  - Copy atmx_flash.FLM and atmx_nvds.FLM to /Applications/SEGGER…./bin/Devices/Atmosic/atmx

2.4.3 Download Firmware and Flash NVDS

After building successfully, select target>Download example, as shown in Figure 4. It will program example.bin and flash_nvds.bin to EVK. Press the PWD button to reboot.

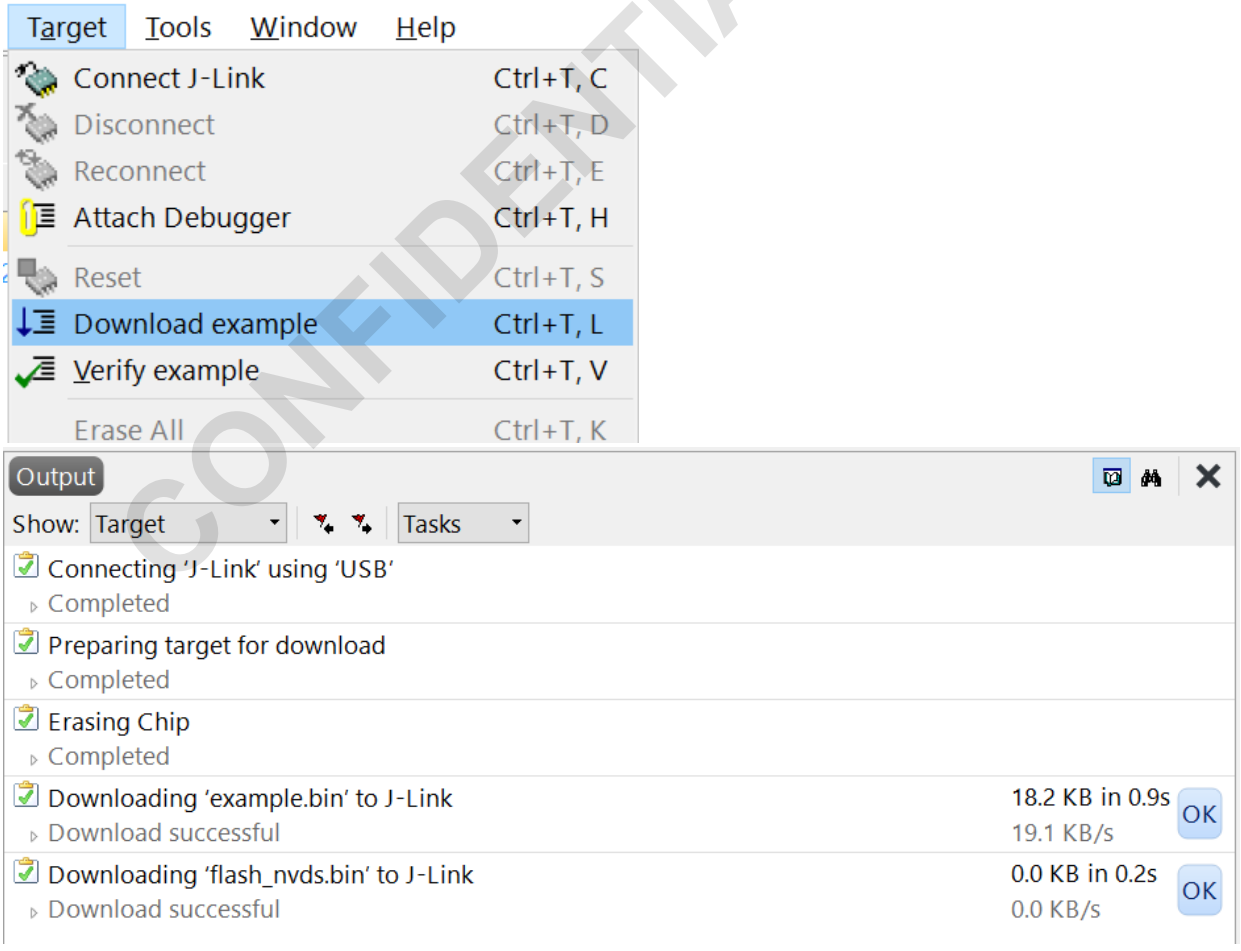


Figure 4 - Programming Flash Output

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2.5 Debug

Before debugging, disable sleep via Flash NVDS (flash_nvds.bin). Please modify tagID (11-SLEEP_ENABLE) and set it's value to 0x00 (Sleep disabled) and then generate the flash_nvds.bin. The SES project is using the flash_nvds.bin of the sesauto folder. Please copy sleep disabled flash_nvds.bin into the sesauto folder. You must set up a J-Link debugging session. SES has an integrated debugger that can be to step through your application. Select Debug>Go

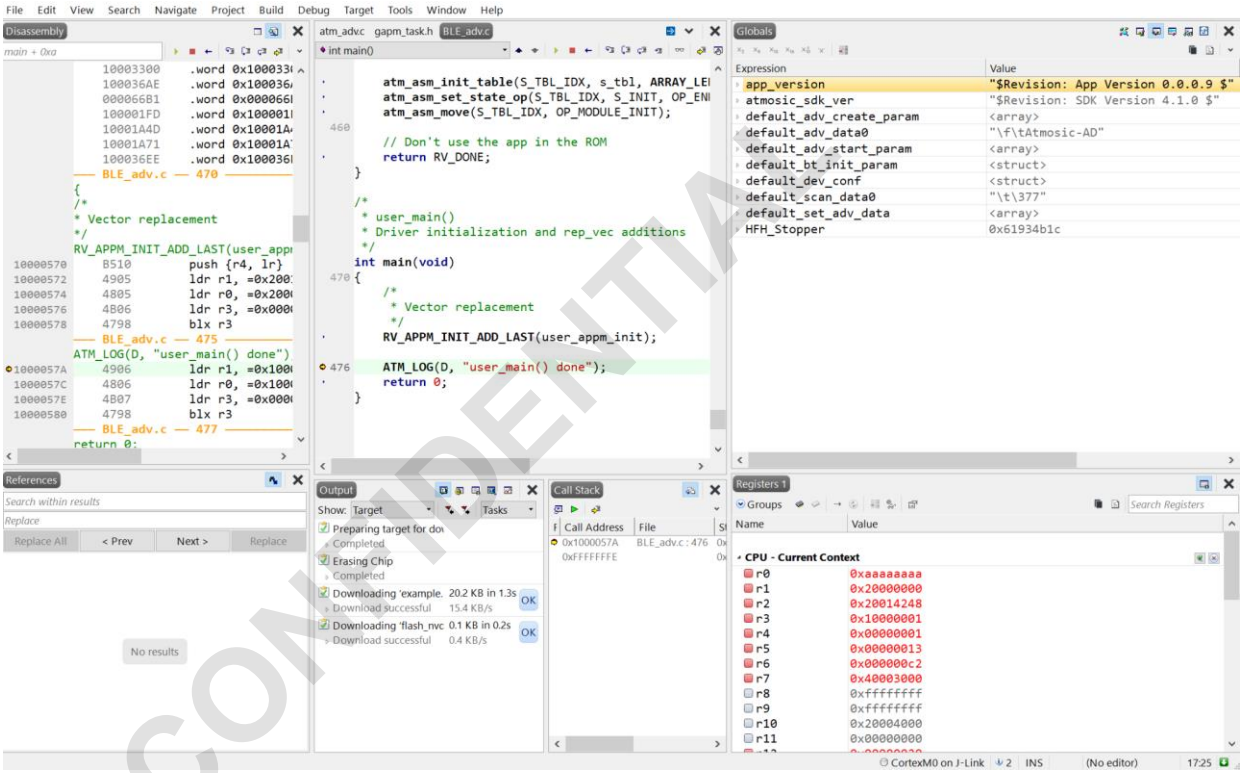


Figure 5 - SES Debug Session Windows

3. ATM33 Platform

3.1 Installation

1) Install Atmosic SDK and follow the REAMDE.Windows in SDK tools folder to setup Universal Serial Bus (USB) device
    a) Connect Micro-B USB cable to ATM3330e/ATM3330 Evaluation Board, see Figure 6. Confirm "JLink CDC UART Port" and “J-Link driver” are showing in Windows Device Manager.

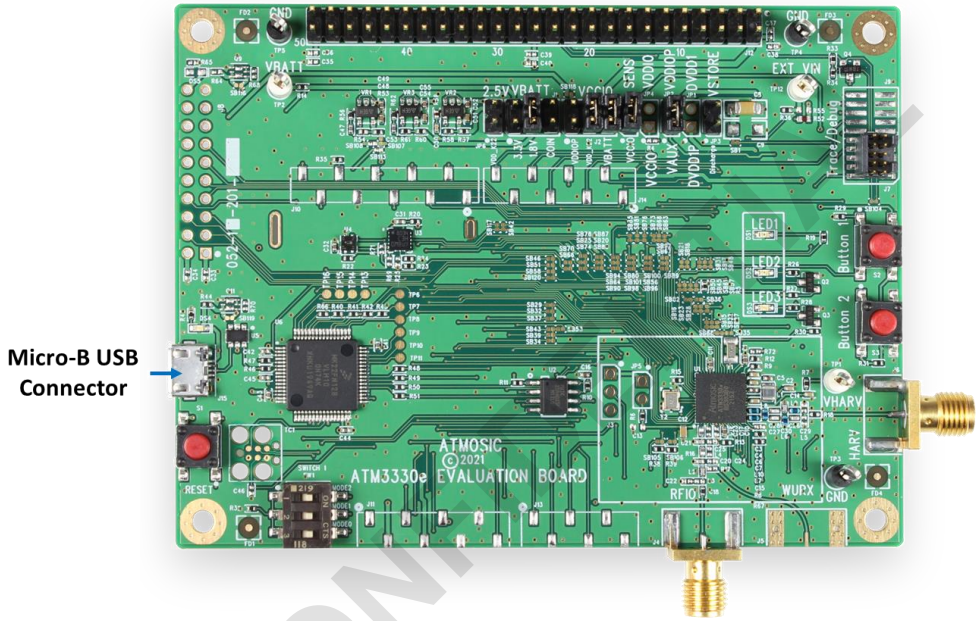


Figure 6 - ATM3330e/ATM3330 Evaluation Board

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For Atmosic SDK Windows Installer users, please go to Windows Start -> Atmosic SDK, run "Setup IDE Environment" and skip step 2-4 below.

2) Install GNU ARM Embedded Toolchain 10.3-2021.07 and add the toolchain path to the user or system path

3) Add a system environment variable: (Replace [SDK installation folder] name with actual installation folder name of Atmosic SDK)

    a) System variable: ATM_SDK_MSYS2

    b) Value: [SDK installation folder]\tools\msys2-base-x86_64-20220128

4) Edit system environment “Path” to add gcc toolchain in path. (Replace [SDK installation folder] with actual folder name)

a) Add “[SDK installation folder]\tools\gcc-arm-none-eabi-10.3-2021.07\bin”

5) Reboot PC for system environment variable update.

6) Install SEGGER Embedded Studio
7) Flash Loader

Copy JLinkDevices.xml from [SDK installation folder]\tools\keil to SES related installed location

Windows 10

a) Copy JLinkDevices.xml to C:\Program Files (x86)\SEGGER\SEGGER Embedded Studio for ARM xxxx\bin\

Linux

b) Copy JLinkDevices.xml to /usr/share/segger_emb…./bin

macOS

c) Copy JLinkDevices.xml to /Applications/SEGGER…./bin

3.2 Generate SES Project File and NVDS

1) Linux and macOS
   a) Navigate to the examples folder (such as **SDK_root\platform\atm33\ATM33xx-3\examples\BLE_adv**)
   b) Enter the command in a command line to generate example code’s SES project: make ses_gen BOARD=m3330 or BOARD=m3330e
   c) You will see a new folder named "ses". This is the SES project file for this example.
2) Windows 10
   a) Each example will have a SES project (folder named "ses") after running the Atmosic Windows SDK installer.
3) NVDS bin file would be generated to ses folder as generating SES project file. The NVDS bin file would not be updated as building examples. To change BD address, please add USER_BD_ADDR:="xx xx xx xx xx xx" to generate SES project. For example, make ses_gen BOARD=m3330 USER_BD_ADDR:="a1 a2 a3 a4 a5 a6". The NVDS file would be program to device as dexecute “Download example”

3.3 Build and Program Examples

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1) Ensure “J-Link driver” is showing in the Windows Device Manager under USB devices.
2) Example project (under ses folder)
3) Edit/Compile/Program firmware in SES IDE, see Figure 7 and Figure 8.

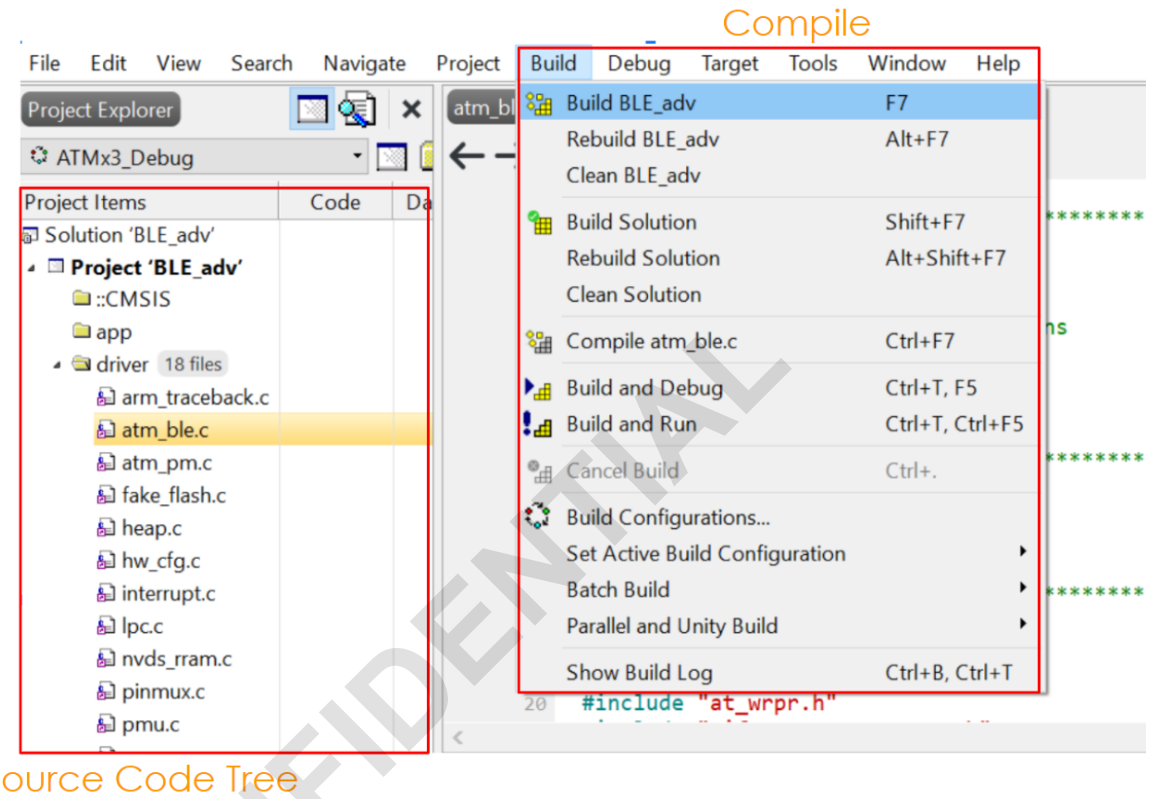


Figure 7 - SES IDE GUI - Compile

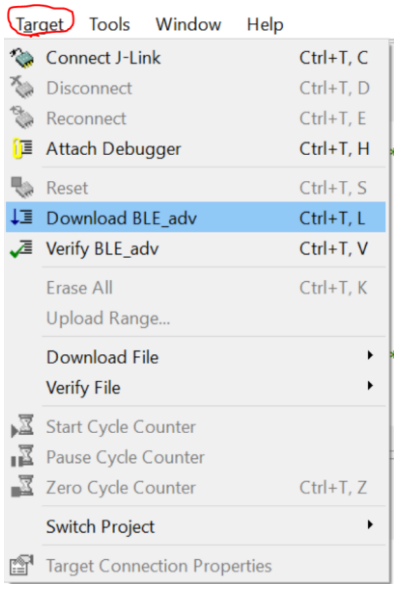


Figure 8 - SES IDE GUI - Program

3.4 Bootloader

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The pre-link process would generate the bootloader binary file. In the SES IDE download process, the bootloader, example firmware and NVDS bin files would be programmed into the device together through J-Link. See Figure 9.

```
> Compiling 'atm_gap.c'
> ../../../../../../tools/gcc-arm-none-eabi-10.3-2021.07/bin/arm-none-eabi-gcc -c -mthumb -march=armv8-m.main+fp+dsp -mcpu=cortex-m33 -mfloat-abi=hard -mfp16-format=ie
> Checking 'atm_gap_param.c'
> Output/ATMx3_Debug/Obj/BLE_adv/atm_gap_param.o does not exist.
> Compiling 'atm_gap_param.c'
> ../../../../../../tools/gcc-arm-none-eabi-10.3-2021.07/bin/arm-none-eabi-gcc -c -mthumb -march=armv8-m.main+fp+dsp -mcpu=cortex-m33 -mfloat-abi=hard -mfp16-format=ie
> Checking 'ble_gap.c'
> Output/ATMx3_Debug/Obj/BLE_adv/ble_gap.o does not exist.
> Compiling 'ble_gap.c'
> ../../../../../../tools/gcc-arm-none-eabi-10.3-2021.07/bin/arm-none-eabi-gcc -c -mthumb -march=armv8-m.main+fp+dsp -mcpu=cortex-m33 -mfloat-abi=hard -mfp16-format=ie
> Checking 'BLE_adv'
> Output/ATMx3_Debug/Exe/BLE_adv.elf does not exist.
> Pre-link command
> "C:/Program Files/SEGGER/SEGGER Embedded Studio for ARM 6.20/bin/emScript" -load C:/Projects/atmosic_sdk/platform/atm33/ATM33xx-3/examples/BLE_adv/ses_BOARD=m3330e/p
> PreBuild ... bootloader and flash nvds
> Call prebuild.bat in Windows system
```

Figure 9 - Build Bootloader

3.5 Debugging

Before debugging, disable sleep via FlashNVDS (flash_nvds.bin). Please refer to the Atmosic SDK guide to modify tagID (11-SLEEP_ENABLE) and set it's value to 0x00 (Sleep disabled) then generate the flash_nvds.bin.

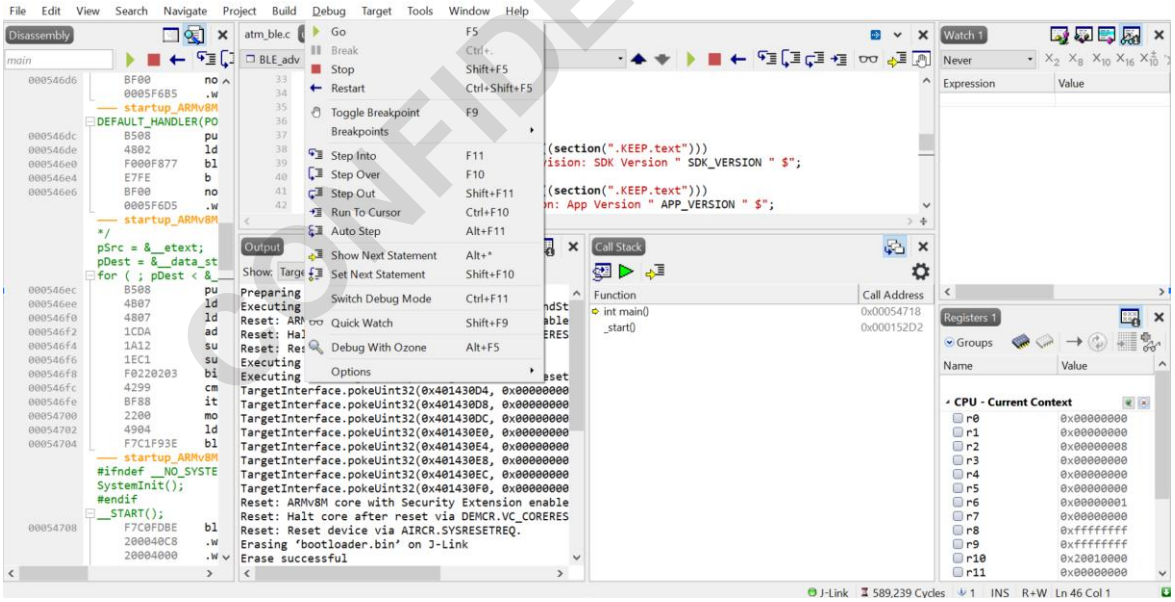


Figure 10 - Debug Mode

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Revision History

Date	Version	Description
June 24, 2022	0.61	Updated Overview item 4),
May 13, 2022	0.60	Initial version created for SDK 5.1 which supports ATM2/ATM3 and ATM33.

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www.atmosic.com